

Examiner reports

Q1.

It was pleasing to see a variety of approaches in part (a), the most popular being to show length of AC = twice length of AD and then to use trigonometry. Many students, however, simply stated AC = twice AD with no justification. Others started from the given result and, after a circular argument, arrived back at the same result. Clearly this does not constitute proof. Students should realise that when two facts are stated (area and length) and a result is expected, both facts are likely to be needed.

The quality of explanation was poor, with great confusion between the use of upper and lower case letters. Errors such as $AB \times BC = AB^2C$, were not uncommon, indicating a lack of experience of working with geometry.

In part (b), most used $\tan A = \frac{\sin A}{\cos A}$, but cancelling of the $\sin A$ meant that only one solution could then be found.

Use of the graph function of the calculator was an acceptable method here, but a sketch and an explanation of what it showed were expected.

For part (c), the majority of students had calculated $A = 60^\circ$ at some point in this question.

Q2.

A significant number of students correctly stated that $\tan 15 = \frac{\sin 15}{\cos 15}$ but, disappointingly, then wrote down the given surd forms of $\cos 15$ divided by $\sin 15$. Many resorted to using their calculator, which gave the final answer immediately, but this approach received no credit as students were clearly asked to fully justify their answer. Some students did not appreciate the necessity of using the conjugate to rationalise.

Q3.

Part (a) was well attempted and many students scored full marks. The question left students with the choice of how to approach the proof and inefficient but correct methods were often seen. Students who scored one mark often failed to complete the proof. Students should be in the habit of finishing any proof or “show that” question with a concluding statement which completes their argument.

While many students scored full marks on part (b) the most efficient method was rarely seen. Students who solved $p(x) = 0$ on their calculators and made use of the factor theorem to write down the correct factorisation could achieve full marks in one line.

Part (c) discriminated well between students of varying abilities. Many students could start the question by manipulating the given equation to form a cubic equation. Some students who went on to see the connection with the earlier part of the question made errors with their use of inequalities or failed to complete their argument with enough rigor.

Some students seemed to focus on the phrase “no real solutions” and were determined to try and use the discriminant of an incorrect quadratic.

Q7.

This final question resulted in a wide spread of marks. In part (a) a significant number of students did not see that the answers could be written down directly. The most common incorrect method led to incorrect equations similar to $x + 30 = 89.27$, the right-hand side coming from $\tan^{-1}(79)$. Most students correctly recognised the geometrical transformation in part (b) to be a translation but the direction of the translation was quite frequently incorrect. In part (c)(i) the right-hand side of the given equation was usually expanded correctly with the correct notation seen and many recalled and used the correct relevant trigonometric identity to eliminate $\sin^2 \theta$. However, solutions towards the printed answer were spoilt in a significant minority of cases by students incorrectly writing the correct expression $5 + 1 - \cos^2 \theta$ as $5 - 5\cos^2 \theta$. Of those who obtained the correct simplified quadratic equation in $\cos \theta$, most were able to solve the equation correctly but not all of these students gave an indication that they had considered and rejected the value -2 for $\cos \theta$. The final part of this last question proved to be a challenge even for the more able

students. Many solutions just consisted of solving $\cos \theta = \frac{3}{4}$ to obtain two solutions rather than stating the required equation as $\cos 2x = \frac{3}{4}$ and solving it to find the four solutions within the given range to the required degree of accuracy.

Q8.

In Part (a), most students gained the method mark by using the identity $\sec^2 x = 1 + \tan^2 x$ and many went on to earn the accuracy marks. A significant number of students lost the final accuracy mark through poor notation such as writing $\tan x^2$ instead of $\tan^2 x$. It was

surprising to see some students working at this level to write $\frac{\sec^2 x}{\sec^2 x - 1}$ as $\frac{\sec^2 x}{\sec^2 x} - \frac{\sec^2 x}{1}$.

Part (b) proved difficult for most students. Many students assumed that they could factorise the quadratic equation and they were satisfied with factors that were clearly incorrect, such as $(\operatorname{cosec} x - 3)(\operatorname{cosec} x + 1)$. Some students misquoted the quadratic formula; some others muddled factors and roots. Most students showed a correct conversion from $\operatorname{cosec} x$ to $\sin x$ after obtaining their roots but those who attempted to convert their equation at the start rarely proceeded to a correct quadratic equation in $\sin x$. Some students discarded the negative root of their quadratic equation.

In part (c), the students who answered part (b) correctly usually gained both marks for this part, although some lost the accuracy mark for not writing their answers to the nearest degree. Where both marks were not earned, the majority of the students gained the method mark for equating $2\theta = 60^\circ$ to one of their solutions in part (b).

Q9.

The majority of students did well on part (a) of this question and well over half of the cohort scored 6 marks or more. Most students were less successful on part (b), although many started by writing down several relevant looking identities.

Part (a)(i). The vast majority of students answered correctly, with relatively few finding $\alpha = 56.3^\circ$ rather than the expected $\alpha = 33.7^\circ$.

Part(a)(ii). Although most students scored the 3 marks some failed to state the minimum

value where an explicit statement was required. Similarly, some students just wrote down the value of x at which the minimum occurred without justifying it. Less common errors were to use $\cos^{-1}\left(-\frac{1}{R}\right)$ or to assume the minimum was at $x = 90^\circ$.

Part (b)(i). Many students wrote the given expression in terms of $\sin x$ and $\cos x$, but then went on to treat the identity as if it were an equation, often equating to zero. Those who used a common denominator or took out a factor of $\frac{1}{\sin x}$ usually completed successfully.

Part (b)(ii). Although most students recognised that they had to use the result from part (i), many failed to consider both $\cot x = 0$ and $\cos 2x = 0$ and so gained no credit. Of those who did consider both these equations many students thought $\cot x = 0$ had no solutions, or they changed it to $\tan x = 0$. Most students did solve $\cos 2x = 0$ correctly, although not always giving both solutions within the given range.

Q10.

Sketching a convincing tangent graph, even with graphical calculators available, seemed a considerable challenge for many. Three separate branches were needed with the correct intercepts on the x -axis and demonstration of the tendency near 90° and 270° . It was disappointing to sometimes see a sine or a cosine graph drawn instead. Many students correctly solved the trigonometric equation in part (a)(ii) although a minority of students incorrectly gave more than two solutions inside the given interval. The majority of students were able to recall and use the required two trigonometric identities in part (b)(i) but not all were able to reach the printed answer without there being a sign error in their working. In the final part of this last question on the paper it was important that students recognised that they were required to use 'Hence'. and also that the equation to be solved was in terms of $3x$. Those students who just solved the equation in part (b)(i) without ever dividing its solutions by 3 or writing ' $\theta = 3x$ ' were heavily penalised. Those students who did link θ to $3x$ generally recognised and solved the quadratic in $\cos 3x$ but a significant minority then failed to find a third solution for x in the given interval.

Q11.

This was a very well answered question with a high proportion of students gaining full marks. A small number of students failed to round results to three significant figures or gave their answers in degrees. Some lost marks through sign errors when collecting the terms of the quadratic together prior to factorisation. The most fundamental error seen was where students tried to write everything in terms of $\tan x$ and occasionally the equation $8(1 + \tan x) - 2(1 + \tan^2 x) = \tan^2 x - 2$ was seen. Those who changed to sine and cosine at the start had to convert to an equation all in cosine to earn marks and some did not go far enough.

Q12.

In part (a), it was clear that many candidates could recall the relevant identity, but rearrangement of $2 \sin \theta = 7 \cos \theta$ into a correct form for applying the identity caused more difficulty than usual.

In part (b)(i), most candidates scored the method mark, but insufficient care in reaching the acceptable form of the printed answer resulted in the loss of the accuracy mark; for

example, answers which involved a mixture of x s and θ s were seen, as well as those which omitted the ' $= 0$ '. In part (b)(ii), almost all candidates realised the need for using part (b)(i) and reached the correct quadratic equation. However, poor factorisation or failing to realise that the roots of the equation after applying the quadratic formula represented the values of $\cos x$, rather than x , led to a significant loss of marks. There were however a substantial number of fully correct solutions which displayed a much better understanding of this type of question than has been the case in similar questions in a number of previous papers.

Q13.

A few candidates made no attempt at this question at all.

Part (a) was very well answered by the majority of candidates, with many obtaining full marks. Most candidates used the correct substitution for $\tan^2 \theta$ and went on to produce the correct quadratic function. Factorisation was usually handled correctly with the common error of

$\sec \theta = -2$ or 5 occasionally seen. The major error noticed was with the answers produced: 60° , 300° , 101.5° were usually correct but the fourth value was often given as 281.5° (from $101.5^\circ + 180^\circ$), rather than the correct solution of $180^\circ + 78.5^\circ = 258.5^\circ$.

In part (b), although most candidates recognised the connection with part (a), not all realised what 'hence' implied. Many gave fully correct answers, but it was disappointing, at this level, how many candidates got the algebra wrong by dividing by 4 first and then adding 10. However, a significant number of candidates chose to start again and most were unable to handle the new expression and gained no further marks.

Q14.

This question, which tested the basic trigonometry section of the specification, gave the value of $\sin \theta$, enabling the area of the triangle in part (a) to be evaluated simply and exactly. This was misinterpreted as a value of the angle θ by some students, for which generous allowance was made in this and the final part. Part (b) specifically asked for the exact value of $\cos \theta$, the significance of which was ignored by many. Those who used the identity $\cos^2 \theta = 1 - \sin^2 \theta$ generally obtained the correct value and full credit was also given for a correct exact value straight from the calculator. The cosine formula was generally well-known and correctly written down in part (c), but there was a substantial minority of students who used the wrong order of evaluation, which resulted in $AC^2 = 30.25 \cos \theta$. A higher proportion of students than usual seemed to have their calculators set in the wrong mode. The final mark was reserved for those calculating the length of AC without unnecessary approximation.

Q15.

This question on trigonometric equations and identities was by far the least-well-answered question on the paper. The equation was given in a factorised form (in the hope that students would be led to write down $\tan \theta + 1 = 0$, $\sin^2 \theta = 3 \cos^2 \theta$) but this was overlooked by many students who chose to multiply out, and create a more complicated problem from which recovery was unusual. Even those who separated the two factors,

$$\frac{\sin^2 \theta}{\cos^2 \theta}$$

and divided the second by $\cos^2 \theta$, often 'cancelled' $\frac{\sin^2 \theta}{\cos^2 \theta}$ erroneously to $\tan \theta$. Of those who overcame the initial hurdles, a substantial number ignored the possibility of a

negative square root and so missed the value $-\sqrt{3}$ for $\tan \theta$.

Q16.

This question was poorly answered.

In part (a), although many students achieved full marks, some found the combining of two fractions beyond them. For many students, an initial step was replacing $1 + \cos \theta$ with $\sin \theta$ or attempting to multiply through by $1 + \cos \theta$, thus obtaining expressions such as

$$1 + \frac{1 + \cos \theta}{1 - \cos \theta} = 32$$

Despite the majority of students finding part (b) of this question difficult, full marks were occasionally seen. Students who were unsuccessful in part (a) were able to attempt this part using the result given. Most students earned the first method mark, but for many this was all they achieved since they only worked with $\sin \theta = +\frac{1}{4}$ and then only worked to 2dp which resulted in just 2 solutions. For the final two B marks, the solution $x = 0.17$ was often missing, and 2.00 was often written as 1.99.

Q17.

Part (a) proved to be a good source of marks, although a significant number of students

lost a mark through poor notation, such as $\cos y^2$ or no $\frac{dx}{dy}$

Although many earned full marks in part (b), many other proofs were inadequate. Students were expected to use the identity $\tan^2 y + 1 = \sec^2 y$ and to show the expansion of $(x - 1)^2$ clearly. Too many fudged their working, wrote $\tan y + 1 = \sec y$ or only proved the known trig identity.

The result in part (c) was easily derived from the previous two parts, but there were many blanks here and many confused statements about inverse functions which were often

taken as $\frac{1}{\tan y}$. Students who used the formula book appropriately generally earned the mark.

There were many fully correct solutions to part (d)(i). However it was distressing at this stage to see some horrendous algebra, with terms inverted in ways that were invalid and worthless.

Only the more able students were able to make progress on the last two parts of the

question. Many students differentiated their quadratic instead of the expression for $\frac{dy}{dx}$. Some students earned the method mark in the final part, but few completed the whole answer, as their final explanation often had an element missing.

Q18.

Part (a) was generally answered very well although some candidates only gave one value within the range despite the clue in the question '....your answers to...' or gave values in the wrong two quadrants.

Part (b)(i), as expected, proved to be the most demanding part question on the paper. The two most common successful initial steps seen were either replacing the 6 on the right-hand side of the equation by $6(\sin^2 \theta + \cos^2 \theta)$ or splitting the $7\sin^2 \theta$ into $\sin^2 \theta + 6\sin^2 \theta$ and then rearranging to get $\sin^2 \theta + \sin \theta \cos \theta - 6(1 - \sin^2 \theta) = 0$. However, for most candidates the maximum mark awarded was 1 for correctly using a relevant trigonometrical identity. Many candidates tried more than one approach, all of which were usually unsuccessful whilst many, having (wrongly) reached the stage $7\tan^2 \theta + \tan \theta = 6$ appeared happy to lose the 7, rearrange and claim the printed result.

It was pleasing to see most of those candidates who attempted part (b)(ii) using the quadratic in $\tan \theta$ printed in part (b)(i) to start their solution. There were many correct solutions presented, although poor factorisation was seen in some attempts. It was surprising to see some candidates using their own version of the quadratic, $7\tan^2 \theta + \tan \theta - 6 = 0$, in part(b)(ii) and this gained no credit.

Q19.

In part (a) most candidates were able to obtain the first angle of 1.77 radians and hence obtain 2 marks. Many candidates also obtained the correct second solution, although 4.91 was a common error.

Part (b) was the worst answered question part on the paper, with very few candidates having any idea of how to go about combining the two fractions. Very few completely correct solutions

were seen. Those who managed to reduce the LHS to $\frac{-2\csc^2 x}{-\cot^2 x}$ could often not complete the question. It was sad to see some very poor algebraic techniques in evidence here. There were many cases where candidates did not attempt the question.

In part (c) far too many candidates failed to recognise both the positive and the negative roots for $\sec x$ and in consequence they did not score any marks, since they only found the solutions for $\cos x = 0.2$.

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Q20.

Part (a) was answered well with most candidates using the trapezium rule with the correct number of strips and values substituted correctly. The most common errors were using degrees instead of radians and incorrect use of brackets.

Most candidates recognised that the required transformation for part (b) was a stretch, but were less sure of the direction and/or scale factor.

In part (c) the most common approach was to attempt to write the given equation in terms of $\tan x$ with varying degrees of success. Although some very good solutions were seen, a common wrong method is illustrated by '2tan $x = 1$, 2 $x = 0.785$, 3.93, $x = 0.393$, 1.97,

Those who correctly reached either $\sin^2 x = \frac{1}{5}$ or $\cos^2 x = \frac{4}{5}$ rarely scored more than one of the

four marks because they forgot to consider both the positive and negative roots of these equations.

Q21.

Expansion of the brackets was tackled in most cases, but with many sign and numerical slips, particularly failing to square the “3”. Most candidates knew they might need to use the identity $\sin^2\theta + \cos^2\theta = 1$, but did not always use it in a constructive fashion. Giving θ a numerical value to find the relevant integer earned no credit in this trigonometrical proof question.

Q22.

In part (a)(i), although the majority of candidates scored both marks here some rounded to the nearest degree, some truncated, some rounded 14.477 to 14.8, some only considered +4, and

some erroneously added -14.5 or 14.5 to 90 . A handful took cosec to be $\frac{1}{\tan}$ or $\frac{1}{\cos}$.

In (a)(ii) there were some excellent full solutions here but also some disturbing misunderstanding of notation. Many candidates found it disconcerting to deal with the functions of $(2x + 30)$, for example $2\operatorname{cosec}^2x - 1 + (2x + 30) = 2 - 7\operatorname{cosec}(2x + 30)$ was seen. They would have been well advised to use a substitution such as Y for $(2x + 30)$. A few missed the brackets around $(1 - \operatorname{cosec}^2Y)$ thus getting a wrong equation and no further marks; some had this correct but failed to deal with the constants correctly and again could not progress. Sadly the incorrect substitution of $\cot Y + 1$ for $\operatorname{cosec} Y$ was also seen on several occasions. Once the correct factors were obtained a few stopped there (particularly those who unfortunately substituted x for $2x + 30$). Quite a few handled the -4 solution well but not the $\frac{1}{2}$ as they went on to assume $\sin(2x + 30)$ was $\frac{1}{2}$ and not 2 .

In part (b) most candidates recognised that the required transformations were a stretch and a translation. The majority started with the stretch and then, wrongly took the translation to be $[-30, 0]$ instead of $[-15, 0]$. A few missed the correct term “stretch” and many failed to use the right terminology of “scale factor” $\frac{1}{2}$. A few had the direction of the stretch in the y -axis or the SF as 2 and a few had their translation in the y -direction.

Q23.

This trigonometry question was the worst answered question on the paper. A common error in part (a) was to either use the wrong identity $\tan(x + 52) = \tan x + \tan 52$ or to solve $x + 52 = \tan^{-1} 22$. Although many candidates scored a mark in part (b)(i) for recalling the

identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$, a significant minority of these candidates could not recall or apply the identity $\sin^2 \theta = 1 - \cos^2 \theta$ correctly.

In part (b)(ii), many candidates applied a correct method to solve the quadratic equation but a significant number of candidates gave two values for $\cos q$ as their final answer, even though the question asked for ‘the value’. It was disappointing to see candidates attempting part (b)(iii), which started with the word ‘Hence’, without linking their solution to

previous work. Although some excellent solutions to this final part were seen, the majority of candidates failed to make any real progress.

Q24.

Part (a) was reasonably well answered, with most candidates obtaining both values. Some candidates lost the accuracy mark through inaccurate evaluation of the second

angle, with 5.94 being a common incorrect answer. A few cases of $\operatorname{cosec} x = \frac{1}{\cos x}$ were seen.

Part (b) was answered very well, with most candidates who obtained full marks in part (a) also obtaining full marks in part (b). The majority of candidates earned the first 4 marks but some then lost the final mark(s) through inaccurate values. There were a few candidates who started with the wrong identity and hence scored zero.

Q25.

Part (a) was well answered with many candidates gaining full marks. Where part marks were earned, the most common error was to lose the factor of 4 in the numerator by incorrectly writing the derivatives of $\sin 4x$ and $\cos 4x$ as $\cos 4x$ and $-\sin 4x$.

Many candidates did not attempt part (b). Although fully correct responses were seen, this was the question part where most candidates scored very few if any marks. Where candidates split the term $4\tan^2 4x$ up into $(4\tan 4x)(\tan 4x)$ or similar expressions they seemed to have more success. In general, the use of the chain rule from whatever starting point was not coped with satisfactorily.

Q26.

This question, which tested trigonometric-graphs, identities and equations, was better answered than similar questions in some recent series. In part (a) most of the graphs presented had the correct shape with intercepts given, although the intercept $y=1$ was occasionally omitted and those on the x -axis only expressed in degrees. Part (b)(i) usually had enough to justify the given result which was then generally used in part (b)(ii). The most common error in the final part was to omit the second answer for x . It was not uncommon to see weaker candidates writing 'identities' such as ' $\cos 2x \times \cos 2x = \cos^2 4x$ '.

Q27.

- (a) Very well answered by the majority of candidates. Most candidates used the correct identity and were successful in answering this part of the question.
- (b) Most candidates were successful in answering this part of the question. When answered correctly the factorisation of the resulting quadratic or use of the quadratic formula was usually well done but there were some sign errors. Many totally correct solutions were seen.

Q28.

Part (a) was well answered by the majority of candidates. $\sec x$ being changed to $\sin x$ was a common error. The first angle was usually found correctly but the second angle was

often incorrect, with 318 being a common error. There were very few cases of angles in radians being found.

In part (b), most candidates used the correct identity and were successful in answering this part of the question. The main error was the use of $\tan^2 x = \sec^2 x + 1$, but $5 \sec x = 5(\tan x + 1)$ was also seen. For those who successfully reached the correct quadratic, the factorisation, or use of the quadratic formula, was usually well done, though there were some sign errors. For the final four answers, many totally correct solutions were seen. A common error at this stage was 284 degrees, from $180 + 104$, and not the correct answer of 256. There were a number of candidates who failed to correctly round their answers to the nearest degree, with $104.47 = 104.5 = 105$ seen.

Q29.

This question which tested trigonometrical identities and solutions of trigonometrical equations was by far the worse answered question on the paper. In part (a) candidates

generally could recall the identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$ but proofs to reach $\tan \theta = 5$ were not always convincing. Those who started with $\sin \theta - \cos \theta = 4 \cos \theta$ were generally more successful than those who started by dividing the numerator by $\cos \theta$.

In part (b)(i) candidates could generally quote the relevant identity, $\cos^2 x + \sin^2 x = 1$, but could rarely use it in a convincing manner.

In the final part of the question, although the majority of candidates could see the use of the work in part (b)(i), a significant number of candidates could not solve the quadratic equation in $\sin x$. In some cases candidates just solved the equation by removing 'sin' wherever it occurred. Better attempts reached $2\sin x = 1$ and $\sin x = -1$ but then it was not uncommon to see ' $2\sin x = 1$ so $2x = 90^\circ$, $x = 45^\circ$, 225° ' and ' $\sin x = -1$ so no solution'.

Q30.

Most candidates obtained the method mark in part (a) by obtaining -0.32 . Many candidates went on to complete this part correctly, although incorrectly writing their answer to 2dp was a common error. Answers in degrees were not common, but they were seen.

Part (b) was answered very well by the majority of candidates with the correct trigonometric identity being used.

In part (c) most candidates were able to successfully factorise the quadratic expression, and many went on to complete the question correctly. Marks were lost by 'extra' values being given or poor accuracy of writing their answers to 2dp.

Q31.

In general candidates found this question difficult, and average-grade candidates rarely scored more than a couple of marks. In part (a), the mark for knowing the correct trigonometrical identity was frequently gained by those who attempted the question, but then many floundered after making the common error " $3 + 1 - \cos^2 \theta = 3 - 3 \cos^2 \theta$ ".

It was encouraging to find a higher proportion of candidates than usual using the printed

answer from part (a), even though they could not obtain it, to start their solution to part (b). Those who realised that θ should be replaced by $3x$ often obtained the 40° solution, but a significant number of these candidates could not then find the remaining two solutions in the given interval.

Q32.

Part (a) was well answered by the majority of candidates. $\tan x = 0.5$ and $x = 0.46, 3.60$ was a common error. The second angle was often incorrect. Few cases of angles in degrees were seen.

In part (b), most candidates used the correct identity and were successful in answering this part of the question. The main error was $\operatorname{cosec}^2 x = \cot^2 x - 1$ followed by fudging of the rest of the solution.

In part (c), most candidates attempted to factorise the quadratic expression, although some used the quadratic formula. Those who factorised were usually correct, although solutions of $\frac{1}{2}$ and -2 were not uncommon. Those who used the formula often made the error of $\cot^2 x = 2$ or $\cot^2 x = -0.5$. Candidates with 3.60 as a solution in (a) were able to recover here but often failed to obtain both marks; 5.17 was a common error.

Q33.

This question, on solving trigonometrical equations, was the worse answered question on the paper. In part (a), although the majority of candidates scored the mark for writing $2x = 48$ (or $x = 24$) many failed to score any of the remaining three marks because they just quoted 156 (from $x = 180 - 24$) as the only other solution.

In part (b) candidates generally quoted the correct identity, $\tan \theta = \frac{\sin \theta}{\cos \theta}$, but a significant number of these candidates were unable to use it to derive the correct value for $\tan \theta$. Those other candidates who tried to solve the equation by initially squaring both sides of the given equation were generally less successful.

Q34.

In general this question was done well by candidates of all abilities. Part (a) was very well answered by the majority of candidates, although $\tan x = 1/3$ and $\sin x = 1/3$ were also seen. 1.23 was usually seen but the second result was often incorrect. There were very few cases of results given in degrees.

Most candidates used the correct identity part (b) in and were successful in answering this part of the question. The main error was candidates using $\tan^2 x = \sec^2 x + 1$ and then fudging the rest of the solution.

In part (c) most candidates attempted to factorise the result from part (b), although some used the quadratic formula. Those who factorised were usually correct, although solutions of $-1/3$ and 1 were not uncommon. For the final 3 answers many totally correct solutions were seen. Candidates with an incorrect solution in part (a) were able to recover here from follow through marks, but often candidates failed to obtain both marks since 0 or 6.18 often accompanied 3.14.

Q35.

Part (a) was very well answered with most candidates gaining all 3 marks but it was not uncommon to see an incorrect quotient formula used (usually the wrong order in the numerator, but sometimes the wrong denominator)

Part (b) was reasonably well answered. However a number of candidates tried to tackle this question by squaring, generally without success.

There were very few successful attempts seen in part (c). Most candidates failed to

provide $\frac{dx}{d\theta}$ at all. Many of those who did make an attempt chose to try a different substitution of their own, either immediately or subsequently.



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